

Research Skills Scale for Senior High School Students: Development and Validation

Emy E. Lacson*, Edilberto A. Dejos Jr.
 For affiliations and correspondence, see the last page.

Abstract

The goal of this paper was to develop and validate a research scale that can measure the research skills of SHS students. From review of relevant literatures, an initial draft of the scale was crafted. The initial draft of the scale, composing of five constructs and 48 indicators, was subjected to validation by a panel of research experts where two items were omitted. The scale was administered to SHS students ($n=126$). Exploratory factor analysis was employed to refine the research instrument and reliability testing was undertaken. As a result, 38 items were identified ($\alpha = 0.968$) in three dimensions: problem conceptualization with 12 items ($\alpha = 0.905$), research methods and data analysis with 18 items ($\alpha = 0.952$) and writing and reporting results with eight items ($\alpha = 0.918$).

Keywords: Research Skills, Scale Development and Validation, Exploratory Factor Analysis

Introduction

The integration of research as a subject or as a learning process is an offshoot in 21st century education. The acquisition of research abilities is now considered as a critical goal in secondary and postsecondary education. As a result, research-related subjects such as Practical Research 1 and 2; Inquiries, Investigations, and Immersions; and Research Projects have been added to the Senior High School (Grade 11 and Grade 12) curriculum to help students develop critical and problem-solving skills through qualitative and quantitative research.

Bower (2010) defines research as an original investigation intended to develop information and insight. As a subject and a method of learning, research aims to be an avenue for students to develop their critical thinking skills, problem-solving skills, and communication skills. It aims to let students create and report knowledge in their different fields of interest and in general making them competent practical researchers and problem-solvers in the community (Fischer et al., 2014).

Students, on the other hand, find research difficult and uninteresting, as has been seen (Kionisala, 2017). Moreover, students are not used to doing research, and, as a result, research outputs are often poor and incomplete. According to the findings of Gilmore and Feldon (2010) and Meerah, et al. (2012), one of the main causes of students' difficulties conducting full research is the lack of preparation in research knowledge and abilities. Meerah, et al. (2012) went on

to say that students are expected to have research abilities, such as the ability to locate and select material, and that they should be prepared in the methodology of conducting research, but this often is not the case. Students lack basic methodological knowledge of formulating and evaluating evidence (Cavagnetto, 2010) and make claims without proper evidence and justifications (Fischer, et al., 2014).

Teachers in senior high schools who are in charge of Practical Research students cannot expect students to provide a research problem at the drop of a hat. Before teaching research subjects, teachers must assess their students' research abilities. Previous studies have shown the value of assessing students' research abilities before they begin conducting research. Meerah et al. (2012) discussed the necessity to craft an instrument to diagnose students' research preparedness. The majority of the accessible research skills scales are designed for undergraduate and graduate students as well as for faculties like, "Measuring graduate students' teaching and research skills" of Gilmore and Feldon (2010) and "Developing an instrument to measure research skills" of Meerah et al. (2012).

As a result, the goal of this research is to develop and validate a measure that can be used to diagnose SHS students' prior research skills. The development of this tool will provide essential information to SHS teachers and curriculum developers in making judgments about fundamental research skills intervention programs for SHS students.

Research Objective

This study aimed to develop and validate a research instrument that can measure the research skills of Senior High School Students.

Theoretical Framework

This study was anchored upon the Research Skill Development (RSD) framework of Willison and O'Regan (2007) and the Conceptualization of Research Skills of Fischer et al. (2014) and Opitz et al. (2017).

Table 1 shows Willison's framework on how students develop research skills and research autonomy. The framework suggested a seven-level continuum development of the 6 facets of research. The six facets of research are Embark and Clarify, Find and Generate, Evaluate and Reflect, Organize and Manage, Analyze and Synthesize, and Communicate and Apply.

According to Willison, these 6 facets are paralleled with 6 affective domains which are the curious, determined, discerning, harmonizing, creative and constructive. The facets are cognitively focused while the affective is for values. This framework articulates what teachers will do to facilitate learning research through active exploration. It is a tool that can be used to analyze and plan curricula for the systematic development of students' research skills. The framework is descriptive, scalable, and useful for assignment, course, or program planning purposes. (Table 1. *Research Skills Development Framework by Willison & O'Regan, 2007* [please see appendix 1])

Furthermore, Opitz et. Al (2017) suggested the conceptualization of research skills as Problem Identification, Questioning, Hypothesis Generation, Construction and Redesign of Artifacts, Evidence Generation, Evidence Evaluation, Drawing Conclusions, and Communicating and Scrutinizing. Opitz et. al (2017) identified these independent skills in newer conceptualizations of scientific reasoning.

Methodology

This study is descriptive research on the development and validation of a research instrument to measure the research skills of SHS students.

Participants

The participants of the study were senior high school

students at an integrated public senior high school during the school year 2021-2022. The study utilized a sample 126 students, purposefully identified for the fruition of the research objective of developing and validating a research skills scale for senior high school students.

Instrument of the Study

The study employed an initial draft researchers-made research skills scale for senior high school students. The five-point Likert scale consisted of 48 items covering five research constructs namely, problem identification and conceptualization skills, information and evidence seeking skills, research methodology skills, statistics/quantitative analysis and evidence evaluation skills, and communication and language skills. The respondents were tasked to indicate their level of agreement with the following responses: (5) certainly true, (4) generally true, (3) somewhat true, (2) generally false, and (1) always false.

Procedures

Development Phase

Review of Relevant Literature

In this phase, an extensive literature review on the studies on basic research skills, development of research skills, as well as on the process of developing research skills scales, was done. Constructs and items were generated also in this phase based on the different literature reviews.

Establishing Face and Content Validity

The draft items were sent to a group of experts (3 SHS research teachers and 2 research experts) for face and content validation. The initial draft was composed of 48 items. Experts were asked to evaluate if the items are essential or not. Results were tallied and content validity ratios (CVR) were computed using the formula, as proposed by Lawshe (1975). The experts were also asked to give comments on the structure and wordings of the items and those were carried out.

Preliminary Scale Pilot Testing

The draft was administered to 10 SHS students online through Google forms to establish administration and content validity. After this phase, the research skill scale was integrated in the Google forms.

Validation Phase

Data Gathering Procedure

To gather the data for the validation phase, the scale was formed using Google forms and the link was sent to the selected GAS SHS Students. One hundred twenty-six students were able to access and answer the scale.

Data Analyses

The main purpose of this research phase was to examine the appropriateness of the items and the internal structure of the constructs that the instrument measures. The following are the different statistical analyses done.

1. Exploratory Factor Analysis (EFA). It is a statistical method that increases the reliability of the scale by identifying the items that can be removed. It is used to identify the dimensionality of constructs or factors by examining the relations between items and factors when the information on the dimensionality is limited. Under EFA, Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy and Bartlett's Test of Sphericity were executed to determine to construct validity and to confirm that the data collected for EFA were appropriate. For KMO, it should be greater than 0.70 according to Dalgaty et al. (2003). While, for Bartlett's Test of Sphericity, it should be statistically significant or $p < 0.05$.

2. Reliability Analysis. It is concerned with the consistency, stability, and dependability of scores. The measure used is Cronbach's alpha for each item. If the alpha value is greater than 0.9, the internal consistency is excellent (Hinton et al. (2004).

Ethical Considerations

Before the conduct of this study, ethical considerations were done. The ethics review form was submitted and approved by the school research committee. After which, the researcher seeks for approval to conduct the data gathering from the school principal and from the parents and guardians of the student-respondent through informed-consent letters.

Results and Discussion

To start the scale development, an extensive literature review was done. Literature in research skills development and measuring research skills was selected. To name, the literature is: Developing an Instrument to Measure Research Skills (Meerah, et al.

2011), Design and Validation of a questionnaire to Measure Research Skills (Alvarado, et al. 2016), Measuring graduate students teaching and research skills through self-report: Descriptive findings and validity evidence (Gilmore & Feldon, 2010), Preliminary Methods and Illustrative Examples in Formulating the Research Frameworks on the Research Writing Process for Senior High School Students (Tabuena 2021), and Research Skills in upper secondary and first year of university (Maddens, et al, 2020), Research skill development spanning higher education: Critiques, curricula, and connections (Willison, 2018). The creation of the constructs and items of the RSQ is based on the works of the authors mentioned here and its alignment also to the Curriculum guide of Practical Research 1 and 2 set by the Department of Education.

Constructs and Item Generation

Based on the literature reviewed, 5 constructs were identified:

1. Problem Identification & Conceptualization Skills
2. Information and Evidence Seeking Skills
3. Research Methodology Skills
4. Statistics/Quantitative Analysis and Evidence Evaluation Skills
5. Communication and Language Skills

Items based on the constructs were also generated. 11 items for the first construct, 10 for the second, 9 for the third, 7 for the fourth, and 11 for the last construct. The initial draft is composed of 48 items.

The instrument is a self-rated questionnaire using a 5-point scale (5- Certainly True; 4- Generally True; 3- Somewhat True; 2-Generally False; 1-Always False)

Then, the drafted scale was subjected to content validation by the identified 5 research experts. Content Validity Ratio (CVR) was computed.

Table 2 below shows the initial draft of the five constructs and items with the CVR. Based on the CVR, 2 items were omitted and were deemed not necessary while 46 were retained. Items, "*I can link research to the real-world affairs.*" and "*I can list all of my references using a specific referencing style.*" were omitted (CVR=0.20) (Lawshe, 1975). (Table 2. *Initial Draft of Research Skills Questionnaire Constructs and Items with CVR* [please see appendix

2)

Validation Phase

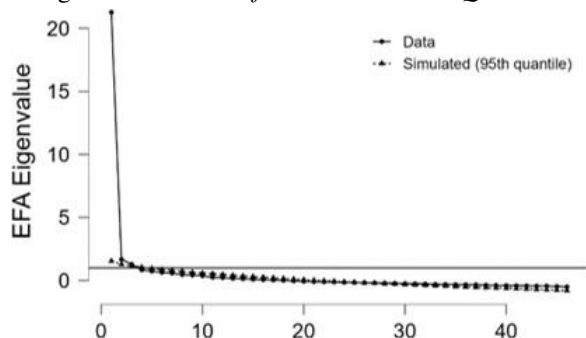
Exploratory Factor Analysis (EFA)

To investigate the structure of the scale, an exploratory factor analysis (EFA) with oblique Promax rotation was carried out using JASP statistic software. EFA is a statistical method used to increase the reliability of the scale by identifying the inappropriate items that can be removed. Descriptive statistics were carried first if no item has a mean closer to 1 or 5. And since no such item was found and the normality of the distribution was confirmed, EFA was carried out.

An initial analysis was run to obtain the eigenvalues for each factor. The Kaiser-Meyer-Olkin (KMO) Sampling Adequacy coefficient which displays the convenience of performing factor analysis for the scale (Dalgaty et al, 2003) was calculated as 0.923, which can be regarded as good. Bartlett's test of sphericity produced a significant result indicating the suitability of the factor model for the data.

Using the factor loadings from EFA, 6 items were omitted because factor loadings lesser than 0.40, 2 items (C5.9 and C5.10) with factor cross-loadings were also omitted and 1 item (C1.10), though with factor cross-loadings was retained but loaded to the factor with higher factor loading.

Figure 2. Scree Plot for Research Skills Questionnaire



Based on the EFA done, three (3) factors emerged contradicting five (5) factors/constructs that were identified from the start of the study. This leads to re-assigning the items and identifying the three factors/constructs of the questionnaire.

Factor 1 is related to the Research Methods and Data Analysis. Factor 2 represents the Writing and Reporting of Results. And lastly, factor 3 appeared to

represent Problem Conceptualization.

Table 3 shows the different items under the 3 factors and their factor loadings. For Research Methods and Data Analysis, 18 items are loaded, 8 items for Writing and Reporting Results, and 12 items are loaded for Problem Conceptualization. (Table 3. *Constructs and Items Factor Loadings* [please see appendix 3])

The Three-Factor Structure Analysis

As shown in table 4, the three factors are highly correlated to each other. The factor correlation between factor 1 (Research Methods and Data Analysis) and factor 2 (Writing and Reporting of Results) was 0.720; the correlation between factor 2 and factor 3 (Problem Conceptualization) was 0.647, and the correlation between factors 1 and 3 was 0.767.

Table 4. Factor Correlation Matrix

	Factor Correlations		
	Factor 1	Factor 2	Factor 3
Factor 1 Research Methods and Data Analysis		0.720	0.767
Factor 2 Writing and Reporting of Results	0.720		0.647
Factor 3 Problem Conceptualization	0.767	0.647	

Reliability Analysis

The reliability of the instrument was established using Cronbach's alpha reliability analyses. Overall, Table 5 shows that the computed value was 0.968 which is regarded as an excellent rating for reliability according to Hinton et al. (2004) as cited by Taherdoost (2016).

Table 5. Overall, Cronbach's Alpha for the Research Skills Scale

Frequentist Scale Reliability Statistics			
Estimate	Cronbach's α	mean	sd
Point estimate	0.968	131.080	24.064
95% CI lower bound	0.960	126.861	21.405
95% CI upper bound	0.976	135.299	27.482

While Table 6 shows that all three factors on the scale had an excellent rating for reliability also. For problem conceptualization, research methods and data analysis, and writing and reporting of results, the computed Cronbach's alpha were 0.905, 0.952, and 0.918

respectively.

Table 6. *Cronbach's Alpha for Each Factor of the Research Skills Scale*

Factor	Cronbach's Alpha	Number of Items
Research Methods and Data Analysis	0.952	18
Writing and Reporting of Results	0.918	8
Problem Conceptualization	0.905	12

As a result of the study, a valid and reliable Research Skills scale was developed by the researcher. Different statistical analyses were performed, and significant results are consistent with those found in the literature and other studies. Though at the start, five constructs were identified through the different related literature but after carrying EFA out, three factors were identified as shown in Figure 3. These are the Problem Conceptualization, Research Methods and Data Analysis, and Writing and Reporting of Results. This is consistent to the proposals of Alvarado, et al (2016) who mentioned the three factors of his research skills scale for engineering students which were: "Process scientific information". "Manage scientific information" and "Develop Scientific information".

Figure 3. *The three-factor research skills model*



Conclusion

This study aimed to develop and validate an instrument to measure Senior High School research skills in the context of their Practical Research subjects. The instrument is called the Research Skills Scale which is consist of 38 items each related to the three identified factors generated from Exploratory Factor Analysis. The three factors are the Problem Conceptualization Skills, Research Methods and Data Analysis Skills and Writing and Reporting of Results Skills.

The instrument has high internal consistency since the Cronbach's alpha coefficient reached the 0.968. Furthermore, correlations between factors indicate also high correlations to each other.

Face and content validity, and reliability have been established, thus achieving the goal of this study. This instrument is a reliable instrument that can be used to measure the research skills of SHS students and can provide important information to SHS teachers and curriculum developers in making judgments about fundamental research skills intervention programs for SHS students.

For future research, it is recommended that this scale be used by research teachers and validation of the study be repeated with other sets of SHS students from other schools. Another recommendation is to conduct a Confirmatory Factor Analysis (CFA) to test predictive validity of the Research Skills Scale. A final suggestion is to extend this study to other significant factors and facets of learning research

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Affiliations and Corresponding Information

Corresponding: Emy Lacson
Email: emy.lacson@deped.gov.ph

Emy E. Lacson
Lapu-Lapu City, Department of Education
Philippines

Edilberto A. Dejos Jr.
Lapu-Lapu City, Department of Education
Philippines

Appendix 1.

Table 1: Researcher Skill Development Framework

A conceptual framework for the explicit, coherent, incremental and cyclic development of the skills associated with researching. © Willison & O'Regan, August 2008/October 2015



www.rsd.edu.au
john.willison@adelaide.edu.au

Researchers...

	← supervisor instigated			↔ researcher instigated	→ discipline leading		
	Prescribed Research Level 1	Bounded Research Level 2	Scaffolded Research Level 3	Self-initiated Research Level 4	Open Research Level 5	Adopted Research Level 6	Enlarging Research Level 7
	Highly structured directions and modelling from supervisor prompt the researcher(s) to...	Boundaries set by and limited directions from supervisor channel the researcher(s) to ...	Scaffolds placed by supervisor enable the researcher(s) to independently ...	Researcher(s) initiate and supervisor guides.	Researcher(s) determine guidelines that are in accord with discipline or context.	Researcher(s) inform others' agendas	Researcher(s) enlarge the field of inquiry.
	Respond to questions/ tasks provided explicitly. Use a provided approach to clarify questions, expectations and ECST issues.	Respond to questions/ tasks implicit in directions. Choose from several provided structures to clarify questions, expectations and ECST issues.	Respond to questions /tasks generated from instructions. Choose from a range of provided structures or approaches to clarify salient elements including ECST issues.	Generate questions/aims/ hypotheses framed within structured guidelines. Anticipate and prepare for ECST issues.	Generate questions/aims/ hypotheses based on experience, expertise and literature. Delve into and prepare for ECST issues.	Identify previously unstated gaps in literature and articulate research directions and ECST issues in response to gaps.	Articulate research directions that expand or direct the field and anticipate the corresponding ECST issues.
	Collect and record required information or data using a prescribed methodology from a prescribed source in which the information/data is clearly evident.	Collect and record required information/data using a prescribed methodology from prescribed source/s in which the information/ data is not clearly evident.	Collect and record required information/data from self-selected sources using one of several prescribed methodologies.	Collect and record self-determined information/ data, choosing an appropriate methodology based on structured guidelines.	Collect and record self-determined information/ data, choosing or devising an appropriate methodology.	Synthesise others' methods to formulate novel methods/ methodologies or apply existing methods to novel applications.	Generate new methods/methodologies that are used widely.
	Evaluate sources/ information/data using simple prescribed criteria to specify credibility and to reflect on the research process.	Evaluate sources/ information/data using a choice of provided criteria to specify credibility and to reflect on the research process.	Evaluate information/data and inquiry process using criteria related to the aims of the inquiry. Reflect insightfully to improve own processes used.	Evaluate information/data and the inquiry process using self-determined criteria developed within structured guidelines. Refines others' processes.	Evaluate information/data and inquiry process using self-generated criteria based on experience, expertise and the literature. Renews others' processes.	Generate substantial research outcomes, so that ideas, practices or interpretations are cited/implemented by others.	Generate substantial research outcomes, so that ideas, practices or interpretations become foundational in field or discipline.
	Organise information/data using prescribed structure. Manage linear process provided (with pre-specified team roles).	Organise information/data using a choice of given structures. Manage a process which has alternative pathways (and specify team roles).	Organise information/data using recommended structures. Manage self-determined processes (including team function) with multiple pathways.	Organise information/data using self or team-determined structures, and manage the processes, within supervisor's parameters.	Organise information/data using self or team-determined structures and management of processes.	Form a research team or a team of community-based practitioners.	Form and develop research networks/communities.
	Interpret given information/data and synthesise knowledge into prescribed formats. Ask <i>emergent question</i> .	Interpret several sources of information/ data and synthesise to integrate knowledge into standard formats. Ask <i>relevant, researchable questions</i> .	Analyse trends in information/data and synthesises to fully integrate components specified. Ask <i>rigorous, researchable questions</i> .	Analyses information/data and synthesizes to fully integrate components, consistent with parameters set. Fill knowledge gaps that are stated by others.	Analyse and create information/data to fill researcher-identified gaps or extend knowledge.	Synthesise others' concepts or interpretations to frame novel outcomes. May also address substantial concerns of a community.	Develop new concepts or interpretations that expand the field or discipline. May also address substantial concerns across communities.
	Use prescribed genre to develop and demonstrate understanding from a specified perspective. Apply to a similar context the knowledge developed. Follow prompts on ECST issues.	Use discipline-specific language and prescribed genre to develop understanding, and demonstrate it to a specified audience. Apply to different contexts the knowledge developed. Clarify ECST issues.	Use discipline-specific language and genres to demonstrate scholarly understanding for a specified audience. Apply the findings to diverse contexts. Specify ECST issues that emerge.	Use appropriate language and genre to address gaps of a self-selected audience. Apply innovatively the knowledge developed to a different context. Probe and specify ECST issues in each relevant context.	Use appropriate language and genre to extend the knowledge of a range of audiences. Apply innovatively the knowledge developed to multiple contexts. Probe and specify ECST issues that emerge broadly.	Change the conversation within the discipline/field through publicly- available communication of knowledge/understanding. Articulate and promote relevant ECST issues.	Change the direction of the conversation across disciplines/ fields. Articulate and promote ECST issues that were previously unstated.

Appendix 2. Initial Draft of Research Skills Questionnaire Constructs and Items with CVR

Research Skills Statements		CVR	Decision
Problem Identification & Conceptualization Skills			
1	If confronted by a question/problem, I can see it as an opportunity to do research.	1	RETAINED
2	I can link research to the real-world affairs.	0.2	Omit
3	I can identify and ask useful, challenging questions; always curious.	1	RETAINED
4	I can formulate my research topic/problem based on related literatures and other sources.	1	RETAINED
5	I can write a research title.	1	RETAINED
6	I can create mind or concept mapping of my research topic/problem.	1	RETAINED
7	I can generate research questions based on the topic/problem.	1	RETAINED
8	I can justify the reasons for conducting the research.	1	RETAINED
9	I can formulate my research hypotheses.	1	RETAINED
10	I can elaborate key variables in my research topic.	1	RETAINED
11	I can indicate the scope and delimitation of my research.	1	RETAINED
Information and Evidence Seeking Skills			
1	I can gather information about my research topic through various means (e.g., electronic media, images, audio and video).	1	RETAINED
2	I can identify and access appropriate bibliographical resources, archives and other sources of relevant information (* including web-based resources, primary sources and repositories)	1	RETAINED
3	I can assess the reliability, reputation, currency, authority and relevance of sources.	1	RETAINED
4	I can evaluate the accurateness of the content by reading other sources mentioned by the writer.	1	RETAINED
5	When searching for information, I can arrange each item systematically.	1	RETAINED
6	I write down the important concepts myself using my own words to support my topic	1	RETAINED
7	I can use the main ideas obtained from the information researched to support my topic.	1	RETAINED
8	I can combine the main ideas from one source or more to form a new idea.	1	RETAINED
9	I can write my references in any citation and referencing formats or styles.	1	RETAINED
10	I follow ethical standards in writing related literature	1	RETAINED
Research Methodology Skills			
1	I can formulate a conceptual framework of my research.	1	RETAINED
2	I can plan and design the research process of a research topic.	1	RETAINED
3	I can determine the appropriate research design or method of my research.	1	RETAINED
4	I understand relevant research methodologies and techniques and their appropriate application within own research area.	1	RETAINED
5	I can justify the principles and experimental techniques used in own research.	1	RETAINED
6	I can select or develop a research instrument to gather necessary data for my research.	1	RETAINED
7	I can determine my appropriate respondents and number of respondents necessary for my research.	1	RETAINED
8	I understand and apply the relevant codes of conduct and guidelines for the ethical conduct of research; seeks advice from supervisor.	1	RETAINED
9	I have basic understanding of legal requirements surrounding research, e.g., Data Protection Act, Freedom of Information Act	1	RETAINED
Statistics/Quantitative Analysis and Evidence Evaluation Skills			
1	I can observe and collect necessary data.	1	RETAINED
2	I can determine which statistical tool or method analysis to use for my research.	1	RETAINED
3	I can perform common statistical tools in any statistical applications like MS Excel, SPSS, Minitab, or others app.	1	RETAINED
4	I can analyze and interpret the results of my statistical treatment or method analysis.	1	RETAINED
5	I can evaluate and systematically organize the data I have gathered.	1	RETAINED
6	I can draw conclusions from patterns and themes gathered from the data.	1	RETAINED
7	I can organize and interpret data using tables and graphs.	1	RETAINED
Communication and Language Skills			
1	I have excellent knowledge of language(s) appropriate for research, including technical language.	1	RETAINED
2	I can understand, interpret, create, and communicate appropriately within an academic context.	1	RETAINED
3	I can prepare grammatically and syntactically correct content for presentations.	1	RETAINED
4	I can communicate research results clearly.	1	RETAINED
5	I can construct my thesis statement clearly.	1	RETAINED
6	I can organize my thoughts and ideas clearly and prepare a manuscript of my research	1	RETAINED
7	I can construct my own conclusion based on the information gathered.	1	RETAINED
8	I can communicate orally the results of my research process.	1	RETAINED
9	I can constructively defend research outcomes	1	RETAINED
10	I can formulate recommendations based on conclusion.	1	RETAINED
11	I can list all my references using a specific referencing style.	0.2	Omit

Appendix 3. *Constructs and Items Factor Loadings*

ITEM	Factor 1	Factor 2	Factor 3
Research Methods and Data Analysis			
I can elaborate key variables in my research topic.	0.426		
I follow ethical standards in writing related literature	0.624		
I can identify and access appropriate bibliographical resources, archives and other sources of relevant information (* including web-based resources, primary sources and repositories)	0.579		
I can assess the reliability, reputation, currency, authority and relevance of sources.	0.645		
I can evaluate the accurateness of the content by reading other sources mentioned by the writer.	0.635		
When searching for information, I can arrange each item systematically.	0.631		
I can write my references in any citation and referencing formats or styles.	0.742		
I can formulate a conceptual framework of my research.	0.615		
I can plan and design the research process of a research topic.	0.578		
I can determine the appropriate research design or method of my research.	0.646		
I can justify the principles and experimental techniques used in own research.	0.587		
I can select or develop a research instrument to gather necessary data for my research.	0.594		
I understand and apply the relevant codes of conduct and guidelines for the ethical conduct of research; seeks advice from supervisor.	0.767		
I have basic understanding of legal requirements surrounding research, e.g., Data Protection Act, Freedom of Information Act	0.531		
I can determine which statistical tool or method analysis to use for my research.	0.625		
I can perform common statistical tools in any statistical applications like MS Excel, SPSS, Minitab, or others app.	0.758		
I can analyze and interpret the results of my statistical treatment or method analysis.	0.642		
I can evaluate and systematically organize the data I have gathered.	0.643		
Writing and Reporting of Results			
I have excellent knowledge of language(s) appropriate for research, including technical language.		0.495	
I can understand, interpret, create, and communicate appropriately within an academic context.		0.575	
I can prepare grammatically and syntactically correct content for presentations.		0.67	
I can communicate research results clearly.		0.804	
I can construct my thesis statement clearly.		0.935	
I can organize my thoughts and ideas clearly and prepare a manuscript of my research		0.772	
I can construct my own conclusion based on the information gathered.		0.636	
I can communicate orally the results of my research process.		0.672	
Problem Conceptualization			
I can indicate the scope and delimitation of my research.			0.657
If confronted by a question/problem, I can see it as an opportunity to do research.			0.504
I can identify and ask useful, challenging questions. I am always curious.			0.476
I can formulate my research topic/problem based on related literatures and other sources.			0.466
I can write a research title.			0.632
I can create mind or concept mapping of my research topic/problem.			0.679
I can generate research questions based on the topic/problem.			0.911
I can justify the reasons for conducting the research.			0.567
I can gather information about my research topic through various means (e.g., electronic media, images, audio and video).			0.597
I can use the main ideas obtained from the information researched to support my topic.			0.443
I can combine the main ideas from one source or more to form a new idea.			0.414
I can observe and collect necessary data that can support my problem.			0.407